

Requirements-driven data preparation for AI-powered systems: A framework and transportation-domain evaluation

Supplementary Material S3

Case Documentation for the Prospective Assessment of the RDP Framework (National Transportation Big-Data Infrastructure)

De-identified summary of System Requirements Specification (SRS) findings and supporting documents

This material substantiates the prospective assessment reported in the manuscript, in which the RDP framework was applied deductively to derive the initiative’s data-related requirements, obligations, and preparation mechanisms. It documents the findings of the System Requirements Specification (SRS) and additional supporting documents that instantiate those derived requirements.

1 Scope

The documentation comprises the SRS findings (Section 2) and two supporting documents (Section 3): a corridor-level proof-of-concept network definition, a national catalogue of traffic-management metrics, and the statement of work for a complementary metropolitan urban traffic-control procurement co-authored by the first author. Section 4 provides the traceability between the derived requirements and the artifacts that instantiate them.

2 Summary of SRS findings

The SRS represents the road network as a smart directed graph rather than a geometric centerline, bridging micro-level tactical control (signal timing, individual lanes, pedestrian safety) and macro-level strategic management (corridors, sub-networks, regions). Its principal findings are summarized below.

2.1 Topological data model

Entity	Definition	Key attributes	Role in measurement
Node	Logical/physical endpoint of a road segment; connectivity anchor	Global UUID; Version ID; type enum; signal-group reference	Stop-Line type enables queue-length and signal-wait measurement; Virtual type subdivides long links; Pedestrian / Cyclist types anchor active-mode safety
Micro-Link	Directed edge between two Nodes with one flow direction and uniform behaviour	Global ID; From-Node ref To-Node ref	Separates movements (e.g., left-turn vs through) for lane-accurate reporting
Connection	Virtual movement inside the intersection box linking an entry to an exit segment	Connection ID; ingress / egress link refs; signal-group ref; movement type	Entity on which crossing time and turn restrictions are defined
Super-Link	Logical aggregation of micro-links and connections forming a continuous corridor	Super-Link ID; route ID & name	End-to-end corridor reporting as one managerial entity
Crosswalk	Pedestrian link crossing vehicle links	Crosswalk ID; pedestrian signal-group ref	Vehicle–pedestrian conflict management (C-ITS)
Sub-Network	User-defined aggregation of corridors / intersections	Member element list; topological-integrity rule	Scope for coordinated and adaptive management
Roadside asset	Signals, signs, VMS/LCS	Modeled as attributes on links via OpenLR linear referencing	“Asset-as-attribute” principle avoids node proliferation

2.2 Common-language standards profile and integration services

Cross-authority exchange is grounded in a national profile of DATEX II (ISO 16157) for data structure and OpenLR for map-agnostic location referencing, supported by a national data dictionary and concept catalogue. A measured-data layer (raw sensor outputs such as occupancy and throughput) is distinguished from an elaborated-data layer (KPIs and levels of service derived by statistical or algorithmic processing). Central integration services include a location-resolution service (OpenLR decoding plus map matching to the national base map), a normalization/standardization service (e.g., field-level NTCIP to inter-center DATEX II), a quality-gate service that attaches a reliability flag, and a cross-authority strategy-synchronization service. Base-map geometry is derived from OpenStreetMap through an engineering transformation that raises resolution and separates road physics from traffic-management logic.

2.3 Derived data-related requirements

Applying the RDP framework to the initiative's objectives produced the data-related requirements summarized below. Clause identifiers are retained from the SRS for traceability.

Clause	Derived requirement (de-identified summary)	Primary obligation
REQ-INT-02	Enforce a data contract per provider at ingestion: schema constraints, semantic completeness, and quality SLOs (e.g., maximum latency); non-conforming records are rejected or tagged.	Accuracy; Completeness; Consistency
REQ-INT-04	Quality gate: every record receives a confidence score after verification of format, completeness, and consistency.	Accuracy; Traceability
REQ-INT-05	Automatically record data lineage and provenance from measurement source through every transformation to the final product, to support explainability and trust for downstream analytics and AI.	Traceability & Provenance
REQ-TOP-04	Maintain topology versions over time to preserve historical data and support model training on past network configurations.	Traceability; Timeliness
REQ-CALC-01	Tag entities lacking data as missing; by default do not impute, because completion is context-dependent for traffic management (a simulation/prediction-based service is deferred).	Completeness (context-dependent)
REQ-CALC-02	Resolve conflicting reports on the same link by context filtering and a weighted average based on quality-gate scores.	Consistency; Accuracy
REQ-CALC-03	Monitor metrics and raise alerts for critical entities using hard, relative (statistical), and trend thresholds.	Consistency; Timeliness
REQ-PUB-02	Distribute to authorized consumers over streaming (push) and REST (pull) channels; structure and metadata based on DATEX II with national extensions.	Confidentiality; Relevance
REQ-BD-01	Retain all data historically in an analytic data lake.	Timeliness; Traceability
REQ-BD-02	Continuously monitor statistical drift in data distributions; trigger recalculation of parameters and retraining of AI models.	Timeliness; Representativeness
REQ-BD-03	Provide tooling to assess representativeness, balance, and coverage, including edge cases, geographic areas, and road-user groups, before training, to prevent bias.	Representativeness
REQ-BD-04 / 05	Provide a secure strategic-investigation environment for modelers; the data lake serves as a continuous substrate for model training.	Confidentiality; Predictive readiness
Privacy retention	Comply with applicable national privacy regulation and GDPR, including retention-policy enforcement, anonymization, and erasure of identifying historical data.	Confidentiality & Identifiability

3 Supporting documents

3.1 Corridor proof-of-concept (worked instantiation)

A signalized urban arterial selected as a proof-of-concept corridor illustrates, at instance level, how the SRS findings are realized.

Layer	Instance-level content
Base infrastructure	OpenStreetMap base, plus a points-of-interest layer carrying static attributes for routine and emergency management: facility type (e.g., a large event venue, an emergency-services station, schools, shelters), capacity, operating hours, and a responsible contact.
Topology	A Super-Link for the arterial; signalized Stop-Line Nodes carrying signal-phase numbers (vehicle and public-transport), active timing plans, and cycle times; Pedestrian Nodes and Links; Connections for in-intersection movements.
Dynamic data	Speed and travel time from probe sources and detectors; volume counts and origin–destination data for all object types (private vehicle, public transport).
KPIs (default per road user)	Private vehicles: level of service (LOS). Signal phases: volume-to-capacity (V/C). Pedestrians: abnormal-red interval. Public transport: ratio of actual to estimated maximum travel time.
Sub-Network definition	Spatial polygon or manual element selection, with a topological-integrity check (selecting two Super-Links automatically includes the connecting Node).
Scenario definition	Flexible time periods with before/after comparison (e.g., before vs after a signal-plan change); at least one anchor (e.g., a specific phase at a critical intersection) and one metric per anchor; measurable targets derived from the priority regime.

3.2 National traffic-metrics catalogue (sample)

The metrics catalogue specifies, per metric, the raw-data inputs, sensing source(s), target road-user group, computation logic, and refresh/baseline cadence. A representative subset is reproduced below; eight of these are formalized acceptance metrics carried into the complementary procurement.

Metric	Road user	Raw-data inputs	Sensing source(s)	Computation / refresh
Speed (spot & average)	Vehicles	Spot and section speeds	Loop detectors; video analytics; probe/FCD; Bluetooth	Mean over interval; per segment / route / sub-network
Travel time	Vehicles, PT	Section entry/exit times; plate matches	Plate recognition; Bluetooth; probe; AVL	Actual vs free-flow estimate; per segment / route
Congestion level (LOS)	Vehicles	Actual and estimated free-flow travel time	Derived	Ratio vs user-defined thresholds
Congestion anomaly	Vehicles	Current vs typical congestion	Derived + historical baseline	Deviation from a baseline recomputed daily, per 15-min slot, per day-of-week, over a rolling multi-month window
Green-cycle utilization	Signal phases	Planned vs actual green per phase	Controller logs	Actual / planned green per phase per cycle
Green-wave efficiency	Vehicles, PT	Actual vs planned route travel time (PT includes dwell)	Controller + AVL	Ratio; per route
Queue length	Vehicles	Speed, position	Video analytics; radar; LiDAR	Instantaneous / average / maximum per lane; “in queue” set by speed and gap thresholds
Throughput	Vehicles	Stop-line crossings	Video detectors	Count crossing the stop line per period

Metric	Road user	Raw-data inputs	Sensing source(s)	Computation / refresh
Pedestrian abnormal-red	Pedestrians	Actual red durations	Controller; video	Mean deviation of red interval from a threshold over consecutive cycles
Pedestrian LOS	Pedestrians	Actual green and cycle time	Controller	Delay-based methodology (HCM)
PT schedule adherence / regularity	Public transport	Planned vs actual arrivals; headways	AVL feed	Deviation from schedule; headway variability vs planned interval
Incident / hazard detection	All	Object and event classification	Video analytics; probe	Location + class (collision, stopped vehicle, debris, pedestrian-in-lane)

4 Traceability of derived requirements to instantiating artifacts

The table below traces each derived requirement, obligation, or mechanism to the RDP layers it implicates (system-level requirement → data-quality dimension → task family) and to the artifact or clause that instantiates it. It substantiates the layer-by-layer mappings reported in the manuscript (Tables 5–7).

Table 1. *Traceability between derived requirements and the artifacts that instantiate them.*

#	Derived requirement / obligation / mechanism	RDP layers (SR → DQ → TF)	Instantiating source
1	Topology-aware national model with stable identifiers (stop-line / virtual / pedestrian nodes, single-behaviour micro-links, connections; national UUID + version ID)	SR1 Reliability; SR11 Interoperability → DQ-3 Consistency; DQ-1 Accuracy → TF-3 Integration; TF-2 Structuring	SRS topology model; corridor PoC
2	Common-language standards profile (DATEX II / ISO 16157 + OpenLR; national dictionary; location-resolution & normalization services)	SR11 Interoperability & Scalability → DQ-3 Consistency → TF-3 Integration	SRS standards profile
3	Data-contract enforcement at ingestion + quality gate with confidence score	SR1 Reliability; SR3 Efficiency → DQ-1 Accuracy; DQ-2 Completeness; DQ-3 Consistency → TF-1 Assessment; TF-2 Structuring	SRS REQ-INT-02 / 04
4	Automatic data lineage and provenance from source through every transformation	SR8 Explainability; SR6 Security / SR7 Privacy → DQ-6 Traceability & Provenance → TF-5 Protection & Governance	SRS REQ-INT-05
5	Context-dependent completeness (missing-data tagging, default no-imputation) + weighted conflict-resolution fusion	SR2 Predictive Performance; SR9 Generalization → DQ-2 Completeness; DQ-3 Consistency → TF-1 Assessment; TF-4 Enrichment	SRS REQ-CALC-01 / 02
6	Drift monitoring + representativeness / balance tooling; context-aligned baselines	SR4 Fairness; SR5 Robustness; SR10 Maintainability → DQ-4 Timeliness; DQ-5 Representativeness → TF-6 Maintenance; TF-1 Assessment	SRS REQ-BD-02 / 03;
7	Topology version management + historical data lake for AI training	SR10 Maintainability; SR2 Predictive Performance → DQ-6 Traceability; DQ-4 Timeliness → TF-5 Governance; TF-6 Maintenance	SRS REQ-TOP-04, REQ-BD-01 / 05
8	Anomaly detection (hard / relative / trend thresholds); fail-safe alerting	SR5 Robustness & Stability → DQ-3 Consistency; DQ-4 Timeliness → TF-6 Maintenance	SRS REQ-CALC-03
9	Privacy compliance (national regulation + GDPR); retention / anonymization / deletion; authorized distribution	SR7 Privacy; SR6 Security → DQ-7 Confidentiality & Identifiability → TF-5 Governance	SRS privacy section; REQ-PUB-02
10	Multi-modal KPI catalogue (LOS, V/C, abnormal-red, PT travel-time ratio) per road user, with sensing source and refresh cadence	SR2 Predictive Performance; SR3 Efficiency; SR4 Fairness → DQ-4 Timeliness; DQ-5 Representativeness; DQ-8 Relevance → TF-4 Enrichment; TF-2 Structuring	Metrics catalogue; corridor PoC